Required Assignment 1.2: Applications of AI

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Q1. AI and competitive advantages

I work in AT&T Communication Services, India Development Center, as part of the AT&T Technology Services organization. My team supports Business Technology Product Development, especially the Sales Tooling ecosystem.

This ecosystem includes platforms such as Salesforce, ROME, ADOPT, SSP, and Business Center. These systems are used to sell, market, order, and manage AT&T products, and they depend on multiple backend systems and engineering teams.

Since these platforms support business-critical sales and ordering flows, even small improvements in engineering delivery, defect analysis, requirements clarity, or production support can improve product launch timelines, sales enablement, and customer experience.

In this context, I see AI as a way to improve competitive advantage beyond automation. The real opportunity is in expanding the reverse engineering scope, improving the software development lifecycle, and reducing go-to-market time. AI can help AT&T engineering teams understand complex existing systems faster, reuse knowledge across teams, and improve the speed and quality of software delivery.

Some practical opportunities are:

- AI can help reverse engineering existing applications, interfaces, workflows, and business rules into reusable BRDs, HLDs, LLDs, interface documents, and operational knowledge.
- A Code Intelligence Factory can organize knowledge across functional areas such as Pricing, Contracting, Credit Check, Accounting, Billing, and Data Storage.
- In areas such as ADI, AVPN, and B-VOIP, business and engineering teams often need to understand pricing rules, ordering flows, contract logic, and backend dependencies. AI can make this knowledge easier to retrieve and reuse.
- AI agents can support business analysis, requirements gap analysis, interface design, quality assurance, production support, and architecture reviews.

- In production support, AI can analyze Azure logs, past incidents, known defect patterns, interface flows, and design documents before escalating the issue to engineering teams.
- Microsoft Teams-based AI assistants can help business leaders and product owners understand current operations and create better future-state plans.
- Project Planner and Program Planner agents can integrate with Jira, SharePoint, and employee skill data to improve planning, resource alignment, and execution.

This aligns well with AT&T's strengths because AT&T already has large-scale platforms, deep engineering talent, rich operational data, and strong domain knowledge across telecom products and business processes.

The challenges will be data quality, access control, security, AI governance, and adoption by engineering teams. Agentic AI-generated outputs must also be reviewed carefully by engineering teams before adoption in real-time projects, especially for architecture, design, production support, and customer-impacting decisions.

For AT&T, AI can create competitive advantage when engineering knowledge becomes easier to find, easier to reuse, and easier to apply across business-critical platforms.

Q2. Aligning AI to Strategy and Organization

AI initiatives should be aligned to business strategy, not started as standalone technology experiments.

At AT&T Technology Services, the current expectation is to deliver business outcomes and engineering deliverables with reduced budgets. Development centers have been asked to operate with nearly 15% lower annual budget compared to the previous year. This creates a clear need to improve engineering productivity, reduce delivery cycle time, and accelerate go-to-market.

At the same time, AT&T has enabled engineers with GitHub Copilot and AI models such as Claude Opus, Claude Sonnet, GPT 5.3, GPT 5.2, and GPT-Mini. The opportunity is to use these tools beyond individual coding assistance and apply them to specific parts of the software development lifecycle.

In my context, I picked AI-powered software development as the main theme. The focus is to apply AI where it can reduce manual effort, improve quality, and support faster delivery.

Key AI use cases identified are:

- Application Interface Design Neo Cortex
- Business Analyst Agent
- Scrum Master Agent
- Quality Assurance Agent
- Azure Logs Analysis Agent
- Grafana Dashboard Builder
- UI Analytics Builder
- Software Quality Enforcer
- Expert Azure Network Topology Reviewer
- Azure Cost / FinOps Optimizer
- Code Intelligence Factory Builder

For example, a Business Analyst Agent can help clarify requirements and identify gaps early. A QA Agent can support test scenario creation and improve coverage. An Azure Logs Analysis Agent can reduce production issue analysis time. A Code Intelligence Factory Builder can help teams understand existing applications and reusable components faster.

This aligns AI directly with AT&T's strategic need to deliver more outcomes with fewer resources. AI should not remain a side experiment or only a coding assistant. It should become part of the software delivery operating model and be measured through productivity, quality, cost, and go-to-market improvement.

Q3. Leadership and AI Advocacy

As an associate director in AT&T, advocating for AI adoption is not about creating excitement around technology. It is about making AI adoption a credible, useful, and low-risk business decision by showing engineering teams and executives how AI can improve real work such as software delivery, production support, and decision-making.

The key steps I would take are:

- Create AI awareness sessions to explain how the latest AI tools are changing software engineering, delivery management, production support, and enterprise technology operations. These sessions should also highlight the risk of not adopting AI where it can create meaningful outcomes.

- Build practical proof-of-concepts. I have worked on a PoC around Model Context Protocol servers and created benchmarks with preferred coding practices. The PoC showed how AI can integrate with commonly used AT&T enterprise tools such as Jira, Azure, Wiki, bug reporting systems, Postgres, and MongoDB.
- Encourage teams to identify AI ideas from current engineering pain points such as repetitive work, delays, quality issues, documentation gaps, and production support effort.
- Evaluate ideas based on business impact, such as reduced cycle time, improved quality, faster analysis, better documentation, or lower operational effort.
- Create a team-level AI roadmap. In my team, we collectively identified 13 AI use cases and organized them into an AI roadmap for the next three quarters.
- Create AI SPOCs for AI discussions, knowledge sharing, and adoption support, so that AI knowledge does not remain with only a few individuals.
- Conduct AI tech sessions to share demos, exchange ideas, and discuss new developments in AI, Agentic AI, and enterprise software engineering.

As AI is evolving fast, leaders must also keep learning. I started a LinkedIn newsletter called AaraMinds, focused on AI adoption, governance, agentic systems, and enterprise transformation: https://www.linkedin.com/newsletters/aara-minds-7454049475859218432/

This helps me organize my learning and share perspectives on how AI is impacting enterprise solutions and leadership.

Q4. Creating Healthy Data and AI Culture

Creating a healthy data and AI culture is important because AI adoption cannot succeed through tools alone. In an enterprise environment like AT&T, teams need the right balance of experimentation, governance, engineering discipline, and business outcome focus.

For my Teams, we have established practical ground rules for both AI tool adoption and AI tool creation.

For AI tool adoption, we encourage AI usage for non-critical but time-consuming engineering activities such as unit test creation, integration test validations, SonarQube code coverage improvement, documentation assistance, and code understanding custom instructions.md using GitHub Copilot.

We have also created reference coding practices (scaffold) so that AI usage remains consistent across engineers and teams. A healthy AI culture should not mean uncontrolled experimentation. It should mean disciplined adoption with proper review.

For AI tool creation, we use a review-based approach:

- Engineers review GenAI demos from different teams and capture observations.
- Another group reviews operational excellence demos and Wiki pages to identify the best reusable practices.
- Senior engineers validate these findings before adoption.
- As an engineering leader with Agentic AI experience, I help convert these learnings into practical AT&T AI use cases and a roadmap linked to business outcomes.

This approach helps AI learning become shared, reviewed, and reusable. It also ensures AI adoption does not depend on a few individuals.

For AT&T, a healthy data and AI culture means teams use AI confidently, verify outputs responsibly, and apply AI to improve engineering productivity, delivery quality, operational efficiency, and business responsiveness.

Q5. Scaling AI in Organizations

A PoC proves possibility, but scaling requires adoption, governance, reusable knowledge, and measurable business value.

In AT&T, scaling AI has practical challenges:

- Business and engineering knowledge is spread across systems, teams, documents, and people.
- Platforms such as Salesforce, ROME, ADOPT, SSP, and Business Center have complex backend dependencies.
- Domains such as Pricing, Contracting, Credit Check, Accounting, Billing, and Customer Provisioning require specialized knowledge.
- Security, compliance, access control, data privacy, and auditability must be built in from the beginning.
- AI solutions built for one team may not scale unless they are designed as reusable capabilities.

Suggested approach is to avoid isolated AI tools for every team. So within, AT&T Technology Services, a better approach would be a shared Engineering Intelligence Ecosystem for the BTPD Sales Tooling namespace.

This ecosystem includes :

- Knowledge from applications, interfaces, design documents, production runbooks, and operational tickets.
- A Code Intelligence Factory across Pricing, Contracting, Credit Check, Accounting, Billing, and Data Storage.
- AI agents for business analysis, interface design, QA, logs analysis, production support, and project planning.
- Integration with Jira, SharePoint, Wiki, employee skill data, engineering repositories, and dashboards.
- Microsoft Teams-based AI assistants for product owners, portfolio managers, and business stakeholders.
- Governance controls access, review, security, auditability, and responsible AI usage.

What makes this effective is reuse. A BRD, HLD, LLD, interface document, test scenario, or production support insight created once should support future projects. The production issue analyzed once should improve future resolution patterns.

For AT&T, successful AI scaling means faster delivery, better engineering quality, reduced production support effort, improved business visibility, and stronger organizational knowledge.